

Android Phone Management System

- Project Presentation
- Using C++ & Data Structures

Introduction

- Simulates mobile phone operations
- Manages contacts, messages, call history
- Console-based C++ application

Objective

- Understand real-world use of data structures
- Improve C++ programming skills
- Implement efficient data handling

Modules

- Contacts → Linked List
- Messages → Queue (FIFO)
- Call History → Stack (LIFO)

System Features

- Add & view contacts
- Send & delete messages
- Make calls & view call logs
- Menu-driven interface

Algorithm

- Start program
- Display menu
- User selects option
- Perform operation using switch-case
- Repeat until exit

Methodology

- Modular design
- Separate functions for each task
- Validation using conditions
- Efficient memory management

Requirements

- Hardware: Computer, Keyboard
- Software: C++ Compiler
- OS: Windows/Linux/Mac

Output

==== ANDROID PHONE SYSTEM ====

1. Add Contact
2. View Contacts
3. Send Message
4. View Messages
5. Delete Message
6. Make Call
7. Call History
8. Delete Last Call
9. Exit

Enter choice: 1

Enter Name: VIGNESH

Enter Number: 9100598298

Contact Added!

==== ANDROID PHONE SYSTEM ====

1. Add Contact
2. View Contacts
3. Send Message
4. View Messages
5. Delete Message
6. Make Call
7. Call History
8. Delete Last Call
9. Exit

Enter choice: 2

VIGNESH - 9100598298

==== ANDROID PHONE SYSTEM ====

1. Add Contact
2. View Contacts
3. Send Message
4. View Messages
5. Delete Message
6. Make Call
7. Call History
8. Delete Last Call
9. Exit

Enter choice: 3

Enter Recipient Name: VIGNESH

Enter Message: HELLO VIGNESH

Message Sent to VIGNESH!

==== ANDROID PHONE SYSTEM ====

1. Add Contact
2. View Contacts
3. Send Message
4. View Messages
5. Delete Message
6. Make Call
7. Call History
8. Delete Last Call
9. Exit

Enter choice: 4

1. To VIGNESH: HELLO VIGNESH

==== ANDROID PHONE SYSTEM ====

1. Add Contact
2. View Contacts
3. Send Message
4. View Messages
5. Delete Message
6. Make Call
7. Call History
8. Delete Last Call
9. Exit

Enter choice: 5

Deleting: To VIGNESH - HELLO VIGNESH

==== ANDROID PHONE SYSTEM ====

1. Add Contact
2. View Contacts
3. Send Message
4. View Messages
5. Delete Message
6. Make Call
7. Call History
8. Delete Last Call
9. Exit

Enter choice: 6

Enter Contact Name: VIGNESH

Calling VIGNESH...

```
==== ANDROID PHONE SYSTEM ====
```

1. Add Contact
2. View Contacts
3. Send Message
4. View Messages
5. Delete Message
6. Make Call
7. Call History
8. Delete Last Call
9. Exit

```
Enter choice: 7
```

```
VIGNESH
```

```
==== ANDROID PHONE SYSTEM ====
```

1. Add Contact
2. View Contacts
3. Send Message
4. View Messages
5. Delete Message
6. Make Call
7. Call History
8. Delete Last Call
9. Exit

```
Enter choice: 8
```

```
Deleting Call: VIGNESH
```

```
==== ANDROID PHONE SYSTEM ====
```

1. Add Contact
2. View Contacts
3. Send Message
4. View Messages
5. Delete Message
6. Make Call
7. Call History
8. Delete Last Call
9. Exit

```
Enter choice: 9
```

```
Exiting...
```

```
I
```

```
=== Code Execution Successful ===
```

Advantages

- Efficient data handling
- Faster operations
- Structured program design
- Easy to use

Conclusion

- Demonstrates practical use of data structures
- Improves problem-solving skills
- Useful academic project

Future Work

- Add GUI interface
- Database storage
- Search & edit contacts
- Security features